

Rotating black holes with AI agents

Reproducing a numerical family
and measuring the shift of extremality

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Course on AI agents for research, Departamento de Física, Universidad de Buenos Aires

Course leaders: Matías Zaldarriaga and Nahuel Mirón-Granese

AI assistance: Claude (Anthropic) and ChatGPT / Codex (OpenAI).

The human authors take responsibility for the scientific claims.

The first prompts

4 September 2026 Original project prompt, excerpt

“Una vez verificados los códigos, el paso siguiente sería emprender la búsqueda de nuevas soluciones.”

Validate the code before exploring new solutions.

Initial scope

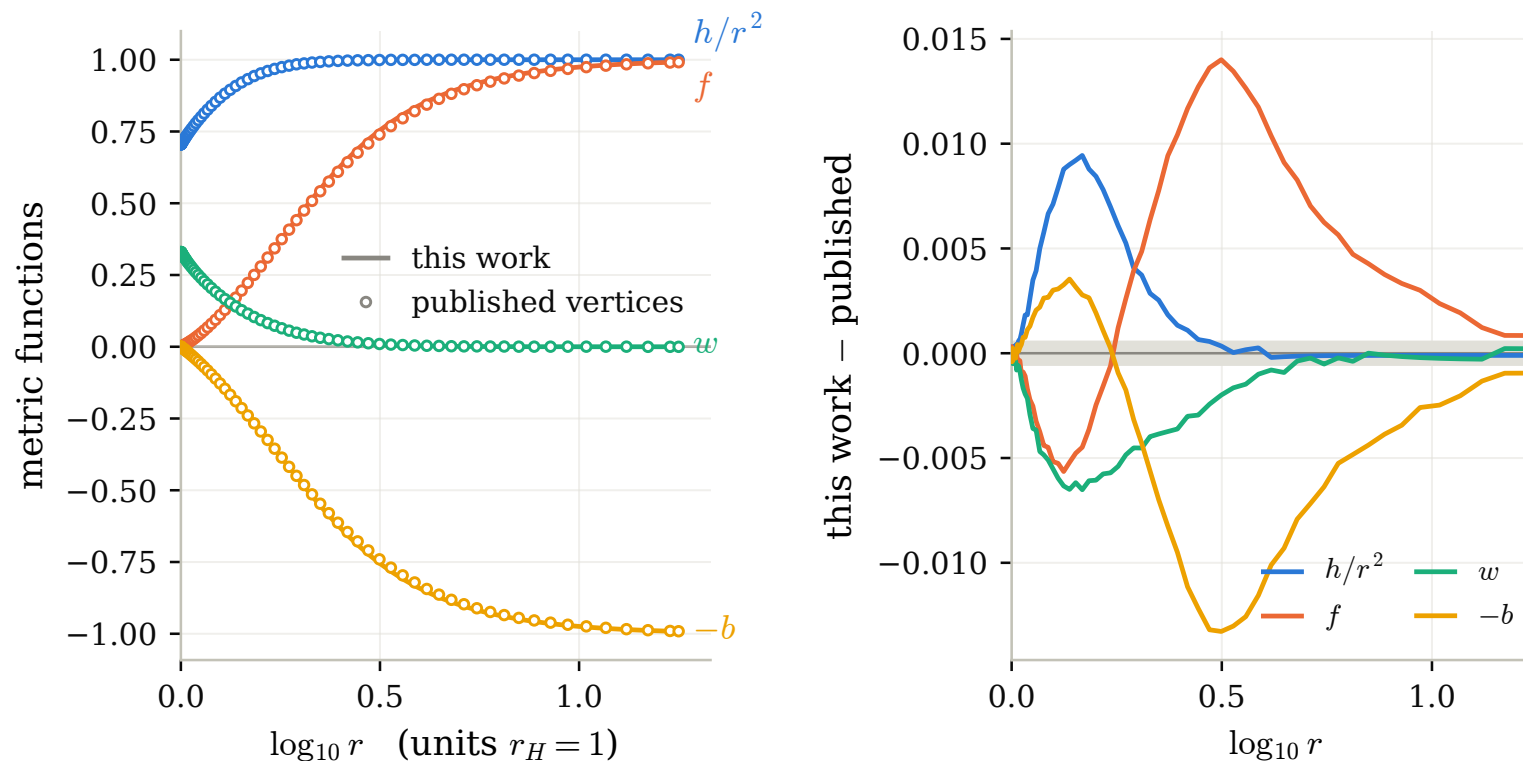
Rotating black holes in $D > 4$
Known GR and EGB benchmarks

The next prompts

Literature list, including 1010.0860
Follow the course requirements

The working problem became $D = 5$, asymptotic flatness and equal spins $J_1 = J_2 = J$.
Source: chronological prompt log, entries 001–003. English line is a paraphrase.

Reproducing the profiles in 1010.0860



Same functions and axes. A visible interior mismatch remains.

$\alpha_{\text{paper}} = 3 = 4\alpha$, $q = 0.33$. Up to 0.014 absolute difference.

74 vertices per published curve, extracted from the source PostScript.

Brihaye, Kleihaus, Kunz and Radu (2010), Fig. 1. Project comparison: short-summary.

Measuring the shift of extremality

Classical Einstein–Gauss–Bonnet gravity in five dimensions, $G_5 = 1$

$$I = \frac{1}{16\pi} \int d^5x \sqrt{-g} [R + \alpha \mathcal{L}_{\text{GB}}]$$

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha \quad \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \lim_{T \rightarrow 0} \Psi$$

Nonlinear solutions

Four radial functions

Chebyshev collocation + Newton

356 states, 13 couplings

Response at each solution

$$R(u, \alpha) = 0$$

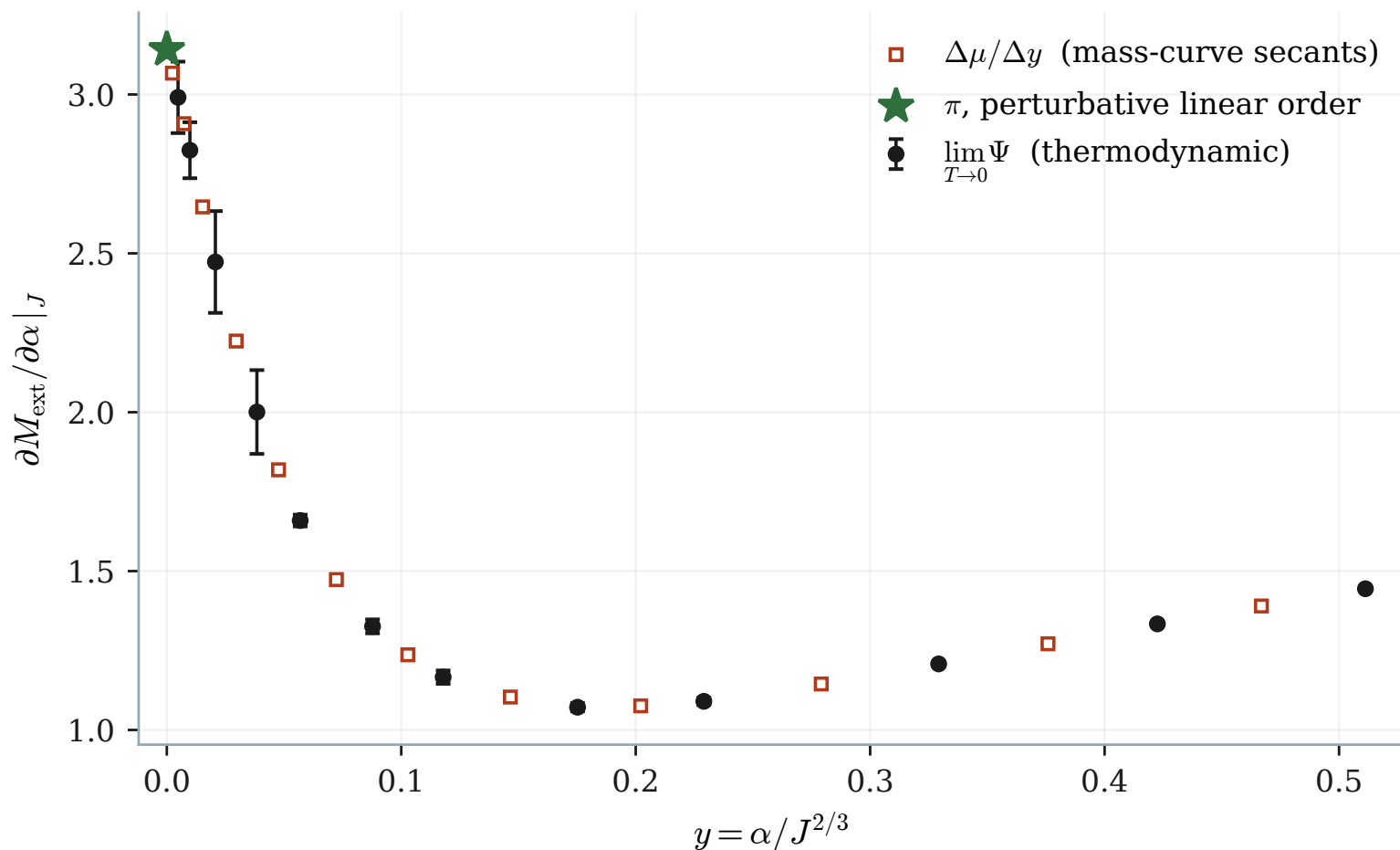
$$R_u u_{,\alpha} = -R_{,\alpha}$$

One linear solve, then extrapolate

Standard first law and numerical methods. Project contribution: finite-coupling response.

The extremal identity assumes a regular branch and finite $\partial_\alpha S_{\text{ext}}|_J$.

The slope stays positive and turns upward



Lowest sample

$$\Psi_{\text{ext}} \simeq 1.07$$

$$y \simeq 0.175$$

Last sample

$$\Psi_{\text{ext}} \simeq 1.44$$

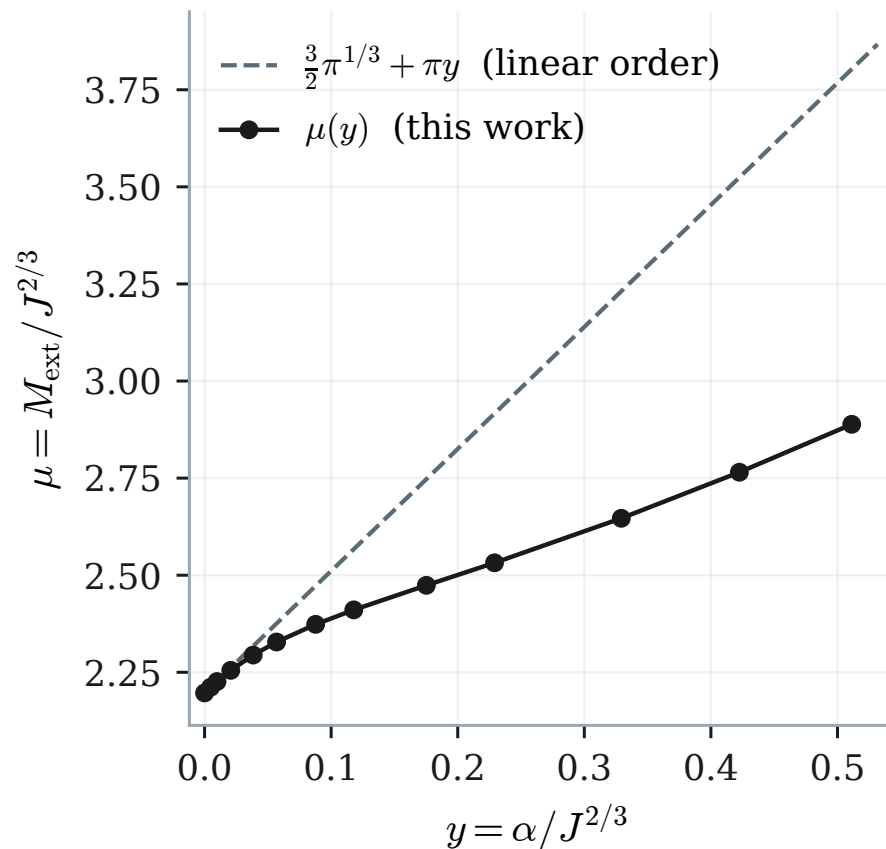
$$y \simeq 0.511$$

Black points: $T = 0$ extrapolants. Bars: fit sensitivity, not confidence intervals.

Star: published first-order slope π (Ma, Li and Lü, 2009.00015).

Open squares: mass secants, which share numerical systematics with the response.

The extremal mass grows more slowly than linear order



About 31% mass growth

Across the sampled range at fixed J

Every extremal point is extrapolated

Branch uniqueness and stability untested

Finite-coupling classical EGB result

No universal higher-curvature EFT claim

Left panel of the manuscript mass figure. $\mu = M_{\text{ext}} / J^{2/3}$ and $y = \alpha / J^{2/3}$.

Recorded time and token use

36.8 h

Active agent time

1.42 billion

Processed tokens

5.94 million

Output tokens

Largest work blocks	Active time	Processed tokens
Rotating EGB equations and solver	11.2 h	508 M
Response, manuscript and review	8.7 h	408 M
First conjugate-potential attempt	4.5 h	169 M

4–28 September 2026. Mostly cached context reads. Counts are not a billing estimate.
Active time uses a 20-minute inactivity cutoff. Human work and unattended numerical runs are excluded.
Total CPU/GPU computation time is unavailable. Source: project usage report.

What we learned about agents in research

What made the result useful

Analytic benchmarks and assertions

Independent checks of the equations

Saved failures, data and run receipts

What still needs human judgment

Choosing a tractable problem

Recognizing shared numerical errors

Limiting claims to the evidence

The physics result is a positive, non-monotonic extremal slope on the sampled branch.

Known family reproduced with explicit discrepancies. Finite-coupling derivative measured in this project.

Next decisive check: independently construct extremal solutions and compare the slope.